

Touch Beyond Reach: Demonstrating Star-Nosed Mole-Inspired Mixed Reality Haptic Proximity Gloves

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Figure 1: "FeltSight" is a bio-inspired system combines custom haptic gloves with a mixed reality headset to extend tactile perception beyond the skin, allowing users to "feel" material textures of surrounding objects without physical contact.

Abstract

FeltSight is an immersive interactive experience that invites participants to inhabit the unique sensory "*umwelt*" of the star-nosed mole. By donning custom-designed, bio-inspired haptic gloves paired with an Apple Vision Pro, users engage in "tele-touch" exploration—feeling the material textures of their surroundings from a distance without physical contact. The demonstration centers on a touch-first interaction model: ten high-precision vibrotactile actuators on each fingertip deliver high-fidelity haptic signatures (such as bark or stone) synthesized from real-world recordings. To facilitate this sensory shift, the XR headset provides a "reduced-reality" visualization, where the environment is hidden until "probed" by the hands, appearing only as transient point clouds that serve as a short-term perceptual memory. Attendees are encouraged to adopt a "tentacular" posture, wandering the space to probe the world

through substituted perception. By inverting the traditional sensory hierarchy and prioritizing proximal touch over distal vision, FeltSight offers a somatic journey into more-than-human awareness, challenging participants to reconsider the boundaries of their own embodiment.

CCS Concepts

• **Human-centered computing** → Interaction design theory, concepts and paradigms; **Mixed / augmented reality**.

Keywords

Mixed Reality, Augmented Humans, Haptics, Sensory Substitution Experience, Vibrotactile Actuators, Star-nosed Mole, Active Sensing, More-Than-Human Design, Tactile Experience, Bio-inspired Interaction, Augmented Virtuality

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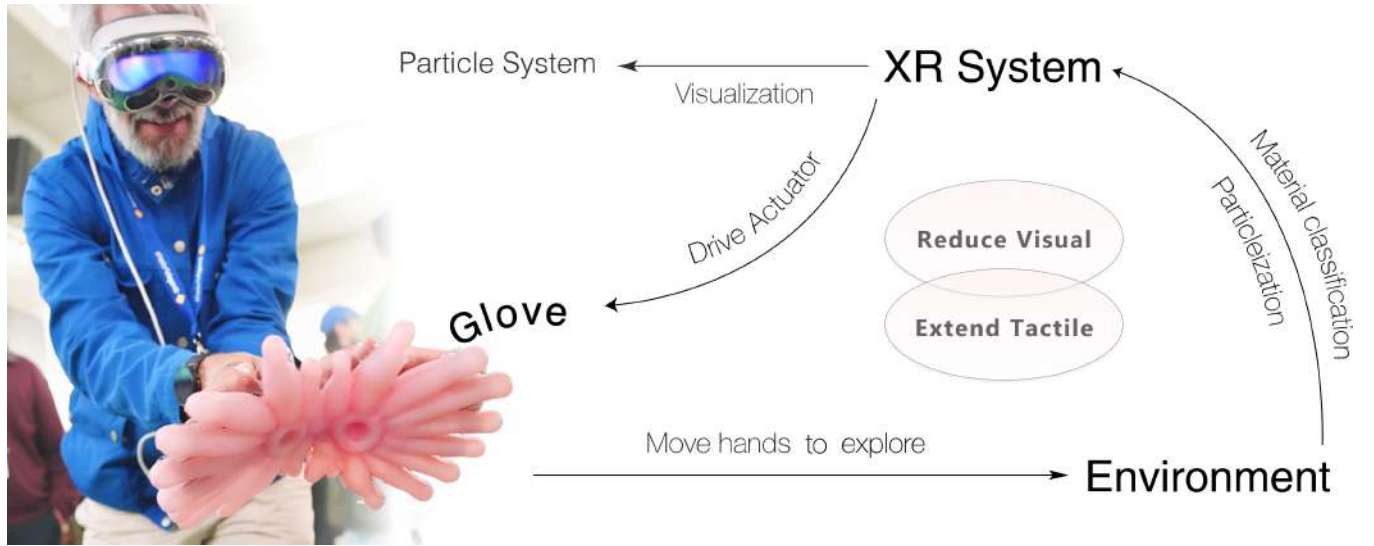


Figure 2: Overview of FeltSight system.

1 Introduction and Related Works

Vision dominates the hierarchy of the human sensory system. This culturally embedded phenomenon of *ocularcentrism* [9] privileges vision as the dominant pathway to knowledge. We propose a departure from this vision-led paradigm by looking toward "more-than-human" sensory models. Specifically, the star-nosed mole (*Condylura cristata*), which is functionally blind, perceives its *umwelt* through 22 fleshy appendages on its snout containing over 100,000 nerve endings [7]. This biological metaphor serves as our design foundation: we invite users to "probe" and "feel" the physical world through a tactile-first interface.

We introduce *FeltSight*, a mixed reality system that recalibrates the user's perceptual hierarchy. By pairing a custom haptic glove with an Apple Vision Pro, we deliberately diminish visual input and amplify tactile sensation. The glove acts as a wearable "snout," transforming the user's hands into a primary sense organ, allowing users to "feel" their surroundings at a distance. This system encourages a "tentacular" way of knowing [6], where the environment is not something merely seen from a distance, but something continuously felt and probed.

Current XR haptic interfaces typically employ kinesthetic force-feedback or electrical muscle stimulation (EMS) to simulate object solidity upon contact [1–5, 10, 11, 13–17]. However, most existing haptic systems only provide tactile feedback during or after contact with virtual content, merely supplementing visual information, regarding touch as the "third sense". *FeltSight* overturns this established and often unquestioned hierarchy by prioritizing tactile sensation as the primary mode of experience. *FeltSight* focuses on pre-contact perception. Leveraging the audio-driven haptic rendering paradigm, our system captures the unique vibration signatures of real-world materials (e.g., bark, stone) and replays them through high-precision vibrotactile actuators on a custom glove [11].

Existing XR haptic research frequently focuses on functional sensory substitution or force-feedback to simulate object solidity

[2, 11, 13]. *FeltSight* moves beyond functional utility, inverting the traditional sensory hierarchy and fostering more-than-human sensibilities [12, 18] with deeper environmental entanglements. By weakening vision, the system nudges participants into a "so-maesthetic" state [8]—a meditative wandering where the pace of movement slows, and the boundary between the physical hand and the digital feedback begins to blur.

2 System Design

The *FeltSight* system shifts the user's sensory hierarchy from distal vision to proximal touch through a custom wearable interface paired with an Apple Vision Pro (AVP) headset. The core hardware consists of a "Mole Glove" cast in soft, flesh-colored silicone elastomer, ergonomically designed to mimic the tactile organ of the star-nosed mole (see Figure 3). This form factor physically guides the wearer into a splayed-finger, "tentacular" posture, forcing a focus on the subtle movements of the hands and the surrounding space. Each fingertip embeds a high-precision vibrotactile actuator (10 per glove), driven by an internal wireless control unit and power amplifiers integrated into the palm structure. This hardware architecture prioritizes a "touch-first" interaction model where the hands serve as the primary organs of perception.

To facilitate this sensory shift, the MR application creates a "visual blackout" environment. To discourage passive reliance on sight, visual feedback is restricted to transient, sparse point clouds that appear only during active probing, functioning as a form of short-term perceptual memory. Interaction is governed by a real-time audio-haptic pipeline: an on-device AI classifies environmental materials at the intersection of finger-projected rays, triggering corresponding high-fidelity haptic textures synthesized from real-world recordings. This feedback is dynamically modulated by the user's motion and proximity to create a "tele-touch" illusion, effectively extending the user's tactile boundary into the environment.



Figure 3: The main glove is cast in soft silicone and shaped to resemble the star-nosed mole's sensory appendages that fan out from around the nostrils, with each finger aligned to a radial tactile "tentacle."

3 Experiencing FeltSight: From Forest to Gallery

To evaluate how FeltSight reshapes human perception and provokes reflection, we deployed the system in two contrasting environments: the forest and the indoor art gallery. These deployments revealed how the system forces a departure from visual dominance toward a primal, tactile-first engagement with the world.

3.1 Workshop in Wild: Rewiring Movement and Mindfulness

During our "in the wild" sessions, participants deviated from linear paths. Their navigation became aimless, nonlinear, and organic—guided by shifting haptic sensations that invited detours toward unseen textures, such as the moss on a fallen log or the subtle ridges of dry mud.

We also observed a transformation in the participants' physical relationship with the environment. Deprived of clear vision, users

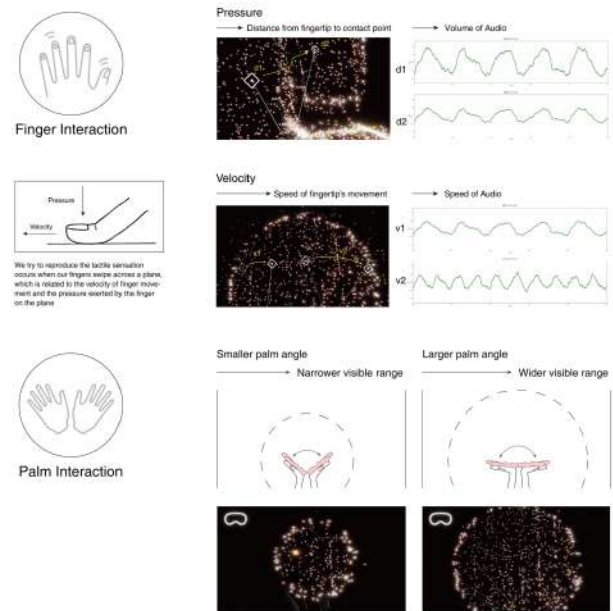


Figure 4: The visible range adjusts based on the angle between the user's palms: a wider angle expands the spherical visible field (up to 1 meter in diameter), while a narrower angle shrinks it (down to 0.5 meters). To simulate the sensation of a finger sliding across material surfaces, we dynamically adjust both audio playback speed and volume of the actuators. Finger movement speed directly controls playback speed, while the distance between finger and object determines the audio volume.

instinctively abandoned an upright, bipedal stance. They naturally began to bend over, and point their gloved hands toward the earth, relying on continuous ground feedback, just like the low-to-the-ground, probing posture of the star-nosed mole. These altered body mechanics suggest a redistribution of sensory priority. This sensory shift doesn't just change information input, it imposes a new *body schema* that prioritizes physical proximity over distant observation.

Beyond physical movement, FeltSight induced a distinct psychological state. A practitioner of walking meditation noted that the system allowed her to enter a "flow state" much faster than usual. By intentionally reducing passive visual input, the participant reported that the system eliminated the cognitive "noise" of the surrounding scenery, leading her attention inward to her senses and thoughts. She reflected: *"I realized that perhaps our overly developed vision sometimes hinders us from entering a flow state."*

3.2 The Gallery: Monstrous Aesthetics and Interface Reflection

In the gallery setting, the experience shifted from private exploration to a form of public performance. The striking, tentacular silhouette of the glove and the unorthodox, mole-like posture of the users created a spectacle that drew significant curiosity. Our design

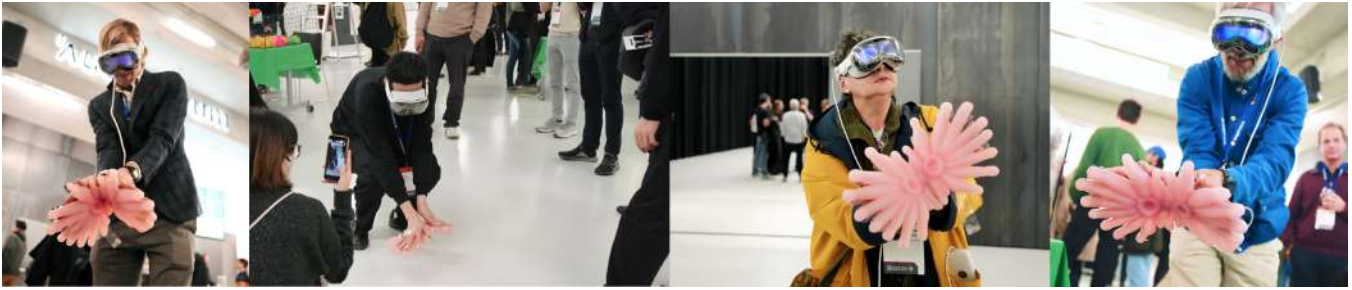


Figure 5: Experiencing FeltSight in the indoor exhibition



Figure 6: Participants often walk with their backs bent, relying on information from the ground.

concept of *Embodying the Monster* proved to be a powerful catalyst for engagement. The "monstrous" experience invited onlookers to reconsider the boundaries of human sensing.

Participants frequently reported a phenomenon akin to the "Rubber Hand Illusion." As the visual point-clouds in the headset synchronized with the vibration of their fingers, users felt as if their proprioception was expanding. One participant remarked that their fingers felt "a meter long," as if they were physically reaching out to touch walls and objects far beyond their natural reach.

Remarkably, the experience prompted a critical reflection on modern digital interaction. A participant poignantly expressed: "While we swipe through digital interfaces every day, we rarely consider that our fingers themselves are the most meaningful interface.... fingertips are high-bandwidth interface, but most digital systems treat them like an on/off button."

4 Conclusion

Inspired by the sensing capabilities of the star-nosed mole, FeltSight introduces a bio-inspired glove and XR interface that inverts the traditional sensory hierarchy by prioritizing touch over vision. By positioning touch as a primary epistemic tool, our system creates opportunities for more-than-human design that respects diverse perceptual experiences. Beyond its theoretical contributions, FeltSight demonstrates significant potential for inclusive and therapeutic applications—ranging from a spatial sensing aid for the visually impaired to a somatic tool for mindfulness and rehabilitation. We demonstrate FeltSight to showcase how sensory technology can serve not merely to augment, but to recalibrate the sensory order—acting as a tuning mechanism to amplify neglected tactile potentials and foster a deeper, embodied attunement to the world.

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A Disclosure of the usage of LLM

We used ChatGPT 5 to facilitate the writing of this manuscript. The usage includes:

- Polish the existing writing.
- Proofread for grammar mistakes and spelling.